

ISyE6669 Homework 8

Fall 2025

1. Consider the following linear program:

$$\begin{array}{llllll}
 \min & -x_1 & - & 2x_2 & & \\
 \text{s.t.} & x_1 & - & x_2 & \leq & 1 \\
 & 2x_1 & - & x_2 & \geq & -1 \\
 & 2x_1 & + & x_2 & \leq & 5 \\
 & x_1 \geq 0, & x_2 \geq 0. & & &
 \end{array}$$

Graph the constraints of this linear program, and indicate the feasible region.

Solution: Please refer to the accompanying figure for the graphical representation of the constraints and feasible region.

2. To solve this problem using the simplex method, first transform it into a standard form LP.

Let x_3, x_4, x_5 be the slack variables with respect to the first, second, third constraint respectively. Denote $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]^\top$ as the vector of variables, and use the standard form notation:

$$\begin{array}{ll}
 \min & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\
 & \mathbf{x} \geq \mathbf{0},
 \end{array}$$

specify $\mathbf{c}, \mathbf{A}, \mathbf{b}$ for the above problem.

Solution: We introduce slack variables x_3, x_4, x_5 to convert inequalities to equalities. The standard form LP becomes:

$$\begin{array}{ll}
 \min & -x_1 - 2x_2 \\
 \text{s.t.} & x_1 - x_2 + x_3 = 1 \\
 & -2x_1 + x_2 + x_4 = 1 \\
 & 2x_1 + x_2 + x_5 = 5 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{array}$$

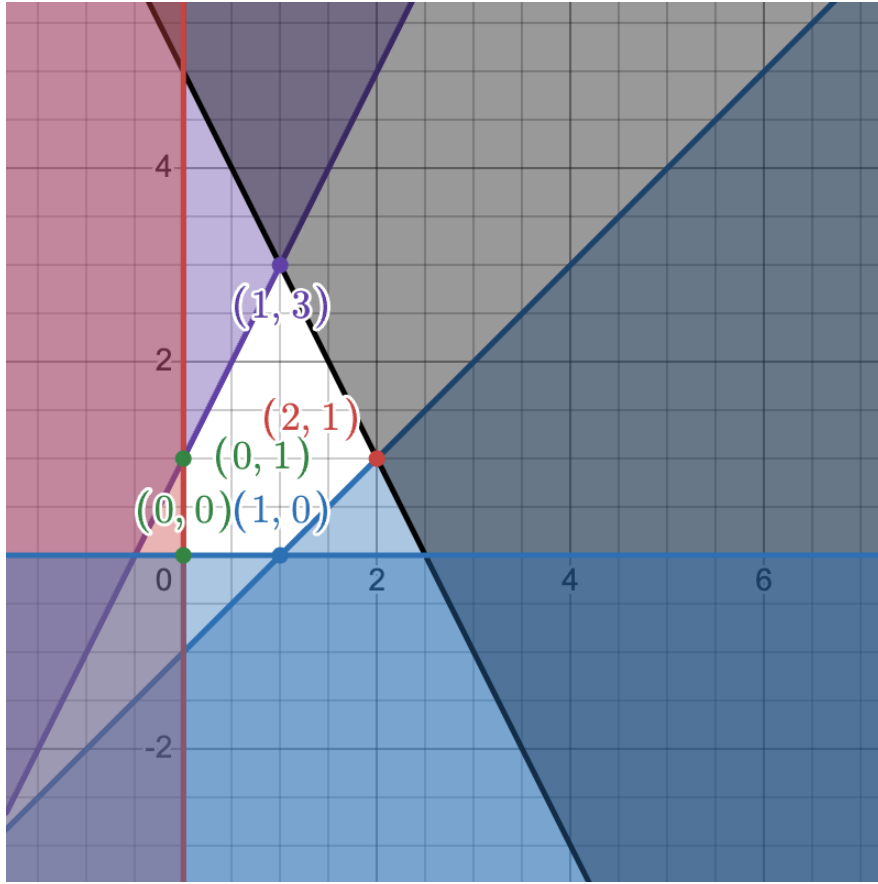


Figure 1: Linear program constraints and feasible region

Therefore, we have:

$$\mathbf{c} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

3. Now we want to solve the above standard-form linear program by the simplex method. If in an iteration of the Simplex method, there is any ambiguity about which nonbasic variable to enter the basis or which basic variable to exit the basis, use Bland's rule.
 - (a) For each iteration of the simplex method, write down the following information in the format given below:
 - Iteration $k = \underline{\text{numerical value, e.g. 1, 2, ...}}$;
 - Basis $\mathbf{B} = [\mathbf{A}_i, \mathbf{A}_j, \mathbf{A}_k] = \underline{\text{numerical value}}$ (i.e. you need to specify i, j, k as well as the numerical values of the columns);
 - Basis inverse $\mathbf{B}^{-1} = \underline{\text{numerical value}}$;

- Basic variable $\mathbf{x}_B = [x_i, x_j, x_k] = \underline{\text{numerical value}}$ (you need to specify i, j, k and numerical values of x_i, x_j, x_k);
- Nonbasic variable $\mathbf{x}_N = [x_p, x_q] = \underline{\text{numerical value}}$ (you need to specify p, q and numerical values of x_p, x_q);
- Reduced cost for each nonbasic variable $\bar{c}_p = c_p - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_p = \underline{\text{numerical values}}$; Same for $\bar{c}_q$; (you need to determine the index “?” for $\mathbf{A}_?$);
- Is the current solution optimal? If not, which nonbasic variable enters the basis?
- Direction $\mathbf{d}_B = -\mathbf{B}^{-1} \mathbf{A}_? = \underline{\text{numerical value}}$. Does the simplex method terminate with an unbounded optimum?
- Min-ratio test $\theta^* = \min_{i: d_{B(i)} < 0} \{x_{B(i)} / (-d_{B(i)})\} = \min\{\underline{\text{numerical values of the ratios}}\} = \underline{\text{numerical value of } \theta^*}$.
- Which basic variable exits the basis?

Start at iteration $k = 1$ with the basis $\mathbf{B}^1 = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5]$. Solve the above linear program with simplex method and write down all the information required above for each iteration. Also indicate the basic feasible solution at each step on the picture in (x_1, x_2) . What is the optimal solution of the above LP in $(x_1, \dots, x_5)$? What is the optimal cost?

From this exercise, you can see how the simplex method works and geometrically what each step is doing.

Solution:

General Framework for Simplex Method:

Let us first establish the general mathematical framework. For a standard form LP with basis matrix $\mathbf{B}$ and nonbasic matrix $\mathbf{N}$, we have:

$$\mathbf{A}\mathbf{x} = \mathbf{B}\mathbf{x}_B + \mathbf{N}\mathbf{x}_N = \mathbf{b} \quad (1)$$

Multiplying by $\mathbf{B}^{-1}$ from the left:

$$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N}\mathbf{x}_N \quad (2)$$

The objective function can be written as:

$$Z = \mathbf{c}_B^\top \mathbf{x}_B + \mathbf{c}_N^\top \mathbf{x}_N \quad (3)$$

Substituting equation (2) into (3):

$$\begin{aligned} Z &= \mathbf{c}_B^\top (\mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N}\mathbf{x}_N) + \mathbf{c}_N^\top \mathbf{x}_N \\ &= \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{N})\mathbf{x}_N \end{aligned} \quad (4)$$

The reduced costs for nonbasic variables are:

$$\bar{\mathbf{c}}_N = \mathbf{c}_N^\top - \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{N} \quad (5)$$

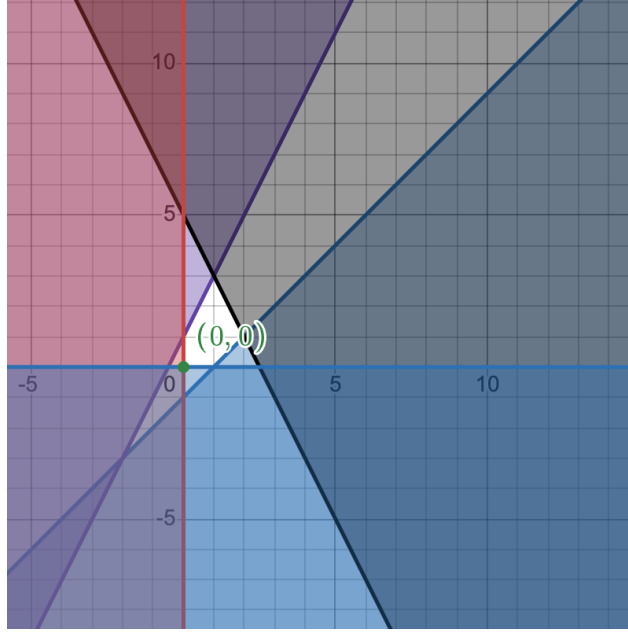


Figure 2: Iteration 1: Basic feasible solution $(x_1, x_2) = (0, 0)$

For entering variable selection, we choose the nonbasic variable with the most negative reduced cost (Bland's rule for tie-breaking). The direction vector is:

$$\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_j \quad (6)$$

where $\mathbf{A}_j$ is the column corresponding to the entering variable.

Iteration $k = 1$:

- Iteration $k = 1$;
- Basis $\mathbf{B}^1 = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$;
- Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$;
- Basic variables $\mathbf{x}_B = [x_3, x_4, x_5] = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$;
- Nonbasic variables $\mathbf{x}_N = [x_1, x_2] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;
- For nonbasic variables, we have $\mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, and $\mathbf{N} = \begin{bmatrix} 1 & -1 \\ -2 & 1 \\ 2 & 1 \end{bmatrix}$.

Using equation (5), the reduced costs are:

$$\begin{aligned}\bar{c}_1 &= c_1 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_1 = -1 - [0 \ 0 \ 0] \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = -1 \\ \bar{c}_2 &= c_2 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_2 = -2 - [0 \ 0 \ 0] \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = -2\end{aligned}$$

- Objective function value using equation (4):

$$Z = \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} + \bar{\mathbf{c}}_N^\top \mathbf{x}_N = [0 \ 0 \ 0] \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} + [-1 \ -2] \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0$$

- The current solution is not optimal since both $\bar{c}_1 = -1 < 0$ and $\bar{c}_2 = -2 < 0$. By Bland's rule, we choose the variable with the smallest subscript among negative reduced costs. Since both are negative, we choose x_1 to enter the basis.

- Direction using equation (6): $\mathbf{d}_B = -\mathbf{B}^{-1} \mathbf{A}_1 = - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}$.

Since $d_{B(2)} = 2 > 0$ and $d_{B(3)} = -2 < 0$, the method does not terminate with unbounded optimum.

- Min-ratio test: $\theta^* = \min_{i: d_{B(i)} < 0} \{x_{B(i)} / (-d_{B(i)})\} = \min\{x_3 / (-d_{B(1)}), x_5 / (-d_{B(3)})\} = \min\{1/1, 5/2\} = \min\{1, 2.5\} = 1$.
- The basic variable x_3 exits the basis (smallest ratio).

Iteration $k = 2$:

- Iteration $k = 2$;

- Basis $\mathbf{B}^2 = [\mathbf{A}_1, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$;

- Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$;

- Basic variables $\mathbf{x}_B = [x_1, x_4, x_5] = \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$;

- Nonbasic variables $\mathbf{x}_N = [x_2, x_3] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;

- For this iteration, $\mathbf{c}_B = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{c}_N = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$, and $\mathbf{N} = \begin{bmatrix} -1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$.

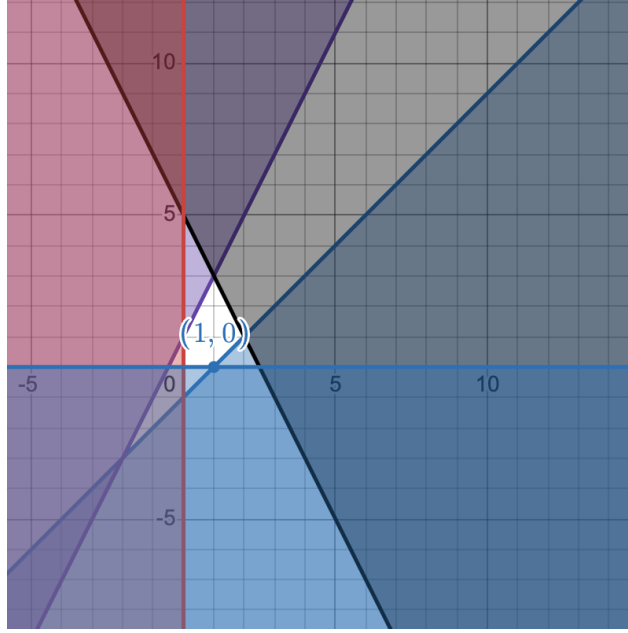


Figure 3: Iteration 2: Basic feasible solution $(x_1, x_2) = (1, 0)$

Using equation (5):

$$\bar{c}_2 = c_2 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_2 = -2 - [-1 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = -2 - (-1)(1) = -3$$

$$\bar{c}_3 = c_3 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_3 = 0 - [-1 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0 - (-1)(1) = 1$$

- Objective function value:

$$Z = \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} = [-1 \ 0 \ 0] \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} = -1$$

- Since $\bar{c}_2 = -3 < 0$, the current solution is not optimal. x_2 enters the basis.

- Direction: $\mathbf{d}_B = -\mathbf{B}^{-1} \mathbf{A}_2 = - \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$.

Since $d_{B(2)} = -3 < 0$, the method does not terminate with unbounded optimum.

- Min-ratio test: $\theta^* = \min_{i: d_{B(i)} < 0} \{x_{B(i)} / (-d_{B(i)})\} = \min\{x_4/3\} = \min\{3/3\} = 1$.
- The basic variable x_4 exits the basis.

Iteration $k = 3$:

- Iteration $k = 3$;
- Basis $\mathbf{B}^3 = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_5] = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}$;
- Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ -4 & -3 & 1 \end{bmatrix}$;
- The basic variable x_5 exits the basis.

Iteration $k = 3$:

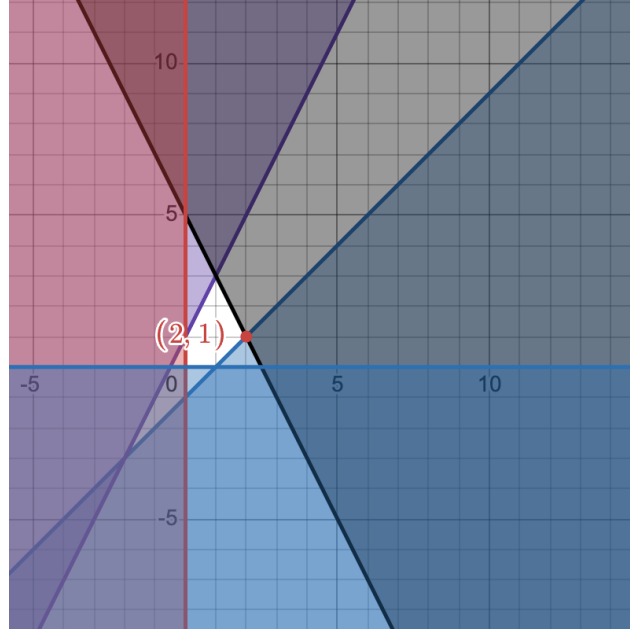


Figure 4: Iteration 3: Basic feasible solution $(x_1, x_2) = (2, 1)$

- Iteration $k = 3$;
- Basis $\mathbf{B}^3 = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4] = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$;
- Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 1/3 & 0 & 1/3 \\ -2/3 & 0 & 1/3 \\ 4/3 & 1 & 1/3 \end{bmatrix}$;
- Basic variables $\mathbf{x}_B = [x_1, x_2, x_4] = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$;
- Nonbasic variables $\mathbf{x}_N = [x_3, x_5] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;

- For this iteration, $\mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$, $\mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and $\mathbf{N} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$.

Using equation (5):

$$\bar{c}_3 = c_3 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_3 = 0 - \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 0 & 1/3 \\ -2/3 & 0 & 1/3 \\ 4/3 & 1 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = -1$$

$$\bar{c}_5 = c_5 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_5 = 0 - \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = 1$$

- Objective function value:

$$Z = \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = -4$$

- Since $\bar{c}_3 = -1 < 0$, the current solution is not optimal. x_3 enters the basis.

- Direction: $\mathbf{d}_B = -\mathbf{B}^{-1} \mathbf{A}_3 = - \begin{bmatrix} 1/3 & 0 & 1/3 \\ -2/3 & 0 & 1/3 \\ 4/3 & 1 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/3 \\ 2/3 \\ -4/3 \end{bmatrix}$.

Since $d_{B(1)} = -1/3 < 0$ and $d_{B(3)} = -4/3 < 0$, the method does not terminate with unbounded optimum.

- Min-ratio test: $\theta^* = \min_{i: d_{B(i)} < 0} \{x_{B(i)} / (-d_{B(i)})\} = \min\{x_1 / (1/3), x_4 / (4/3)\} = \min\{2 \times 3, 4 \times 3/4\} = \min\{6, 3\} = 3$.
- The basic variable x_4 exits the basis.

Iteration $k = 4$:

- Iteration $k = 4$;

- Basis $\mathbf{B}^4 = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3] = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 1 & 0 \\ 2 & 1 & 0 \end{bmatrix}$;

- Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 0 & -1/4 & 1/4 \\ 0 & 1/2 & 1/2 \\ 1 & 3/4 & 1/4 \end{bmatrix}$;

- Basic variables $\mathbf{x}_B = [x_1, x_2, x_3] = \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$;

- Nonbasic variables $\mathbf{x}_N = [x_4, x_5] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;

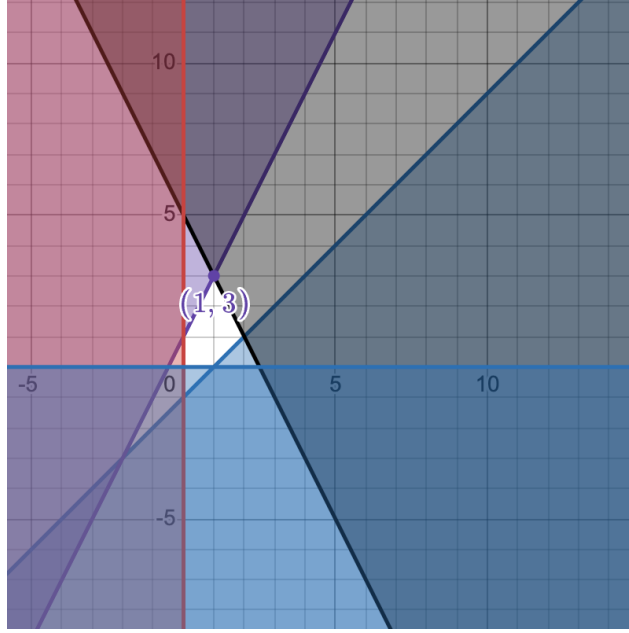


Figure 5: Iteration 4: Basic feasible solution $(x_1, x_2) = (1, 3)$ - Optimal

- For this iteration, $\mathbf{c}_B = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$, $\mathbf{c}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and $\mathbf{N} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Using equation (5):

$$\begin{aligned} \bar{c}_4 &= c_4 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_4 = 0 - \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1/2 \\ 3/4 \end{bmatrix} = 1 \\ \bar{c}_5 &= c_5 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_5 = 0 - \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/2 \\ 1/4 \end{bmatrix} = 5/4 \end{aligned}$$

- Objective function value:

$$Z = \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} = -7$$

- Since both reduced costs are non-negative ($\bar{c}_4 = 1 > 0$ and $\bar{c}_5 = 5/4 > 0$), the current solution is optimal.

Summary: The optimal solution is $\mathbf{x}^* = [1, 3, 3, 0, 0]^\top$ with optimal objective value $Z^* = -7$.

In terms of the original variables (x_1, x_2) , the optimal solution is $(x_1^*, x_2^*) = (1, 3)$ with optimal cost $-1 \times 1 - 2 \times 3 = -7$.

The basic feasible solutions visited during the Simplex method correspond to:

- Iteration 1: $(x_1, x_2) = (0, 0)$ with cost 0
- Iteration 2: $(x_1, x_2) = (1, 0)$ with cost -1
- Iteration 3: $(x_1, x_2) = (2, 1)$ with cost -4
- Iteration 4: $(x_1, x_2) = (1, 3)$ with cost -7 (optimal)